# Andrew McKnight

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## Skills

- *Mobile*: Swift, Objective-C++, SwiftUI, Core Data, AVFoundation, CloudKit, Xcode Instruments
- *Backend*: Vapor, Fly.io, Supabase, AWS, Django, Rails, Google Cloud Run/BigTable
- *Tools*: Claude Code, SPM, XcodeGen, SwiftLint, clang-format, pre-commit, Fastlane, Arkana
- *Infrastructure*: GitHub Actions, TestFlight, Linear, Sentry

## Experience

**Sentry** · Senior Software Engineer · Remote **Nov 2021 – Current**

- *Deliverables*
  - Seer for iOS and macOS: proposed and built native apps to manage our agentic offerings
  - Sentry Cocoa SDK: app monitoring SDK for Sentry (200K orgs, 1M+ developers)
    - Integrated Specto's profiler into Sentry's Cocoa SDK and iterated with new API/features
    - Built User Feedback widget, the first user-facing UI shipped in Sentry Cocoa SDK
    - Parallelized release CI job from 35 to 5; split test suites to fix statefulness issues
    - Built demo apps for SDK and Sales teams flexible enough to simulate any customer configuration
  - Billing: platformized legacy system for product teams to self-serve how they charge for features

**Specto** · Senior Software Engineer · Remote **Nov 2019 – Nov 2021**

- *Highlight*: First employee at startup and remained through acquisition by Sentry
- *Deliverable*: Specto SDK, mobile app profiling SDK for iOS and Android
  - Shared C++ core between iOS and Android with Protobuf RPC to Python/Go backend
  - Hardened SDK init sequence, app launch profiling, NSURLProtocol hooks, MetricKit ingestion
  - Python scripts to generate scoped Xcode projects from Bazel targets (pre `rules-xcode`)
  - Smoke test generation validating integration across CocoaPods, Carthage, Bazel and SPM
  - Signal and mach exception propagation with open source contributions to other crash reporters

**Insight Therapeutics** · Lead iOS Developer · Remote · Contract (Part-time)
**Nov 2011 – Jun 2019**

- *Highlights*
  - Sole consultation and development of multiple apps shipped in App Store and internally
  - Led client-server design with backend engineer, and UX design with clinical admins
- *Deliverables*
  - eConsent: HIPAA-compliant informed consent app for clinical trial participants
    - Internal deployment to hundreds of facilities and thousands of study subjects
    - Secure admin mode for saving/resuming progress and managing completed consents
    - Quiz-based presentation of study details with participant signature collection and export
  - Antibiotic Kinetics: antibiotic dosage calculator
    - Ported from partner developer's Android app codebase, including adding first tests
    - Improved UX for error reporting through multistep calculations and data entry
  - DementiaRx: public health reference e-pamphlet about dementia
    - First new app created by me shipped to the App Store for a contracting engagement
    - Scripted automatic content extraction from original paper publication PDFs
    - Built search, bookmarking and interactive patient eval elements later backported to ClotRx
  - ClotRx: public health reference e-pamphlet about anticoagulants
    - Preexisting hybrid app with webviews and a PDF viewer, modernized by me
    - Migrated UIWebView→WKWebView, added search and bookmarking, redesigned for dark mode
    - Built interactive clinical patient eval modules with print/email reporting capability

**121nexus** · Software Engineer · Remote · Contract (Part-time) **Jan – Apr 2018**

- *Deliverable*: 121 Scan, medical inventory management app/SDK with custom barcode/QR scanning
  - Optimized critical barcode scanning component for high throughput pass/fail operation
  - Traced silent scan failures to an AVFoundation defect with double-FNC1 GS1 barcodes
  - Public launches via App Store and a white-labeled enterprise deployment in logistics facilities

**Layer** · Software Engineer · Remote **Dec 2016 – Jul 2017**

- *Deliverable*: four-layer SDK offering for data→widgets→prebuilt module→sample app
  - Migrated network protocol with Erlang backend from SPDY to WebSockets under Thrift RPC
  - Rearchitected auth and transport layers as state machines using TransitionKit
  - Separated unit, integration and end-to-end tests to tighten development iteration loops
  - Streamlined developer environment configuration (Rake, Gems, CocoaPods and git submodules)

**Twitter** · Software Engineer · Cambridge, MA **Apr 2015 – Dec 2016**

- *Highlights*
  - Presented NSOperation rearchitectures and open source tools at MobileDevCon 2016 in Amsterdam
  - Worked on high-volume lambda backend, including on-call and migration from Storm to Heron
- *Deliverables*
  - Fabric SDK (deployed to 2+ billion devices) and macOS app: developer tool ecosystem management
    - Rearchitected Apple SDK launch sequence using queues of async/compound NSOperations
    - Streamlined build and release processes with TeamCity agents deployed via Chef
  - Crashlytics SDK: crash reporting and beta deployment SDK for Apple platforms and Android
    - Researched and built feature surfacing solutions to detectable ecosystem-wide crash causes
    - Added Bitcode and tvOS compatibility to SDKs, and investigated watchOS support
    - Rearchitected CLIs to bootstrap projects, upload dSYMs and interop with our macOS app

**Raizlabs** · Software Engineer · Boston, MA **Jul 2013 – Dec 2014**

- *Deliverables*
  - Care.com: service seeker/provider marketplace with over 14 million members
    - iOS 7 redesign, with custom perspective effect in UIPageViewController with CATransform3D
    - Added floating quick-post button, swipeable UITableViewRowAction, UIDataDetectorTypes
    - Improved pull-to-refresh, data entry with inputAccessoryView, UIAlertView presentation/routing
  - B&H Photo: e-commerce app for a high end brick-and-mortar merchant
    - Helped rebuild B&H's iPhone app on the new iPad codebase as a universal iOS 7 app
    - Push notifications and deep linking via Appboy, crash reporting and breadcrumbs via Crittercism
    - Custom parallactic scrolling of table view headers, and barcode scanning for product search
    - Custom UICollectionView layouts and animated transitions between grid and row layouts
  - UniFirst: interactive enterprise sales catalog on iPad
    - Improved drag-and-drop clothes modeler and emblem designer built with Auto Layout and KVO
    - Generated sales proposal PDFs from Mustache HTML templates, with model/emblem snapshots
    - Added Core Data model deep copying and orphan cleanup, enabling editing existing sales orders
  - Invaluable: realtime luxury auction platform on iPad
    - Refactored critical bidding calculation routines and corrected mathematical errors
    - Added pull-to-request footer in infinite scroll UITableView, horizontal UICollectionView scroll
    - Fixed statefulness issues with UI elements dependent due to race conditions in server links

## Education

**Old Dominion University** · M.Sc., Computer Science **2013**